Ulas Kamaci

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Summary

PhD in Electrical and Computer Engineering (UIUC) with 10 years of research experience in machine learning for computational imaging and inverse problems. Thesis develops deep learning reconstruction, diffusion models, and calibrated uncertainty quantification for a nonlinear imaging inverse problem. On NASA's ICON mission, built the image artifact-removal pipeline and data-quality flags used in the public data releases.

Education

University of Illinois Urbana-Champaign (UIUC) — Urbana, IL
PhD in Electrical and Computer Engineering — Defended Jul 2026
Thesis: *Inverse Theoretic Approaches for Slitless Spectral Imaging*
Advisor: Prof. Farzad Kamalabadi

Middle East Technical University (METU) — Ankara, Turkey
BS in Electrical and Electronics Engineering — Jun 2017

Skills

Machine Learning	Generative Models (Diffusion Models, GANs), Bayesian Inference, Uncertainty Quantification (Deep Ensembles, Conformal Prediction), Transformers, U-Nets/CNNs, Optimization (ADMM/ISTA)
Core Tools	Python, PyTorch, NumPy, SciPy, Git, Linux
Also Used	Keras, Scikit-learn, Scikit-image, MATLAB, C/C++
Domains	Computational Imaging, Computer Vision, Signal Processing, Remote Sensing, Fourier Optics

Experience

Graduate Researcher — Aug 2017 – Aug 2026
Coordinated Science Laboratory, University of Illinois Urbana-Champaign — *Urbana, IL*

PhD Thesis Research Python, PyTorch, Scikit-image — github.com/ukamaci/slitless

- Constructed a training dataset from ∼2,400 Hinode/EIS observations (2006–2025): per-pixel spectral fits as ground truth, simulated instrument measurements, quality masking, and leakage-free splits, yielding 23k training patches.
- Trained a **U-Net** that recovers intensity, velocity, and line-width maps from overlapping multi-order measurements, **halving** velocity error vs. the classical iterative solver at **60x** faster inference.
- Showed the U-Net with 2 detectors beats classical methods with 5, pointing to cheaper instrument designs.
- Developed a **conditional diffusion model** that matches the U-Net's accuracy while returning posterior samples; it cut velocity error **2.4–8.8x** vs. diffusion posterior sampling and sampled **27x** faster on the same GPU.
- Built a **deep ensemble** of 8 conditional diffusion models for **2-12x** better-calibrated uncertainty quantification than any single model, and conformalized it for interval coverage guarantees.

ICON: NASA Explorer Mission PyTorch, Keras, NetCDF — github.com/UIUC-SINE/icon-fuv

- Owned the **Level 2.5** product of ICON's Far Ultraviolet imager, the nighttime ionospheric density retrieval, as its point of contact; added its quality flags and delivered each release to the mission's Science Data Center through 2025.
- Built the imager's **star and hot-pixel removal** for its calibrated (L1) data around a residual **U-Net** trained on 50k synthetic star-injected frames; used in the public data releases.
- Developed the L1 **uncertainty model**, estimating detector gain from flight data to separate photon shot noise from background noise; its 1-sigma errors weight the L2.5 retrieval and propagate into its published error bars.
- Served 3 months as ICON **Tohban** (on-duty mission monitor): reviewed science data and telemetry daily, **approved data for public release**, and coordinated reprocessing with data product leads.

MAS: NASA Small Spacecraft Technology Program Python, Keras, Scikit-image — github.com/UIUC-SINE/MAS

- Optimized **sensor placement** for a high-resolution **diffractive imaging system** via subset selection, improving reconstruction **SSIM by 32%** over the conventional focal-plane design.
- Combined a **plug-and-play ADMM** solver with a learned CNN denoiser (DnCNN) for **multi-frame deconvolution**, improving reconstruction **PSNR by 2.4 dB** (SSIM 0.63 to 0.70) over a transform-learning baseline.

VISORS: NSF Space Weather Research Program Python, NumPy, Scikit-image github.com/UIUC-SINE/MAS

- Built an **imaging simulator** for formation-flying spacecraft with a diffractive lens, modeling the optical PSF with Fourier optics, sub-pixel shifts between frames, motion blur from linear drift, and detector read, dark-current, and Poisson noise.
- Designed an efficient **multi-frame image registration** algorithm and a plug-and-play ADMM reconstruction with a BM3D denoiser for highly noisy, motion-blurred measurements.

Research Mentoring

- **Mentored 3 students** (2 undergraduate, 1 M.S.) on diffusion-based reconstruction, unrolled networks, and distribution shift, guiding each from problem formulation through experiments and write-up; co-authored a BASP 2025 poster.

Teaching Assistant Jan 2021 – May 2026
Department of Electrical and Computer Engineering, UIUC *Urbana, IL*

- Taught 7 semesters across 5 courses, from graduate pattern recognition and digital imaging to ML programming and DSP; delivered graduate lectures on deep networks, diffusion models, VAEs, and optimization.

Summer Intern Jul 2016 – Aug 2016
Aselsan Research Center *Ankara, Turkey*

- Reformulated patch-based **sparsifying transform learning** for **MRI** reconstruction as an ADMM algorithm and derived closed-form solutions to every subproblem.
- Coded the solver from scratch in MATLAB, improving PSNR over the original formulation at 5x undersampling.

Selected Projects

- **GAN Data Augmentation for Brain Tumor Segmentation** (Computer Vision course, UIUC, Dec 2021; PyTorch). Augmented U-Net brain tumor **segmentation** (BraTS MRI) with **DCGAN**-synthesized image/mask pairs, reaching a **92.3** Dice score vs. 88.5 on real data alone.
- **FRC-GAN: Resolution-Aware Super-Resolution** (CS598PS, UIUC, Dec 2020; PyTorch). Designed a GAN loss from **Fourier ring correlation** for 4x super-resolution; SRGAN trained with it tracked ground truth better at every spatial frequency and lost its fake textures.

Publications

- **Diffusion Posterior Sampling for Slitless Spectral Imaging** 2026
  U. Kamaci and F. Kamalabadi. *IEEE ICASSP*. [link]
- **Solar Slitless Spectroscopy with Learned Reconstruction** 2026
  U. Kamaci and F. Kamalabadi. *The Astrophysical Journal*. [link]
- **In-Flight Performance of the Far Ultraviolet Instrument (FUV) on ICON** 2023
  H. U. Frey, S. B. Mende, R. R. Meier, **U. Kamaci**, et al. *Space Science Reviews*. [link]
- **Optimal Measurement Configuration in Computational Diffractive Imaging** 2020
  E. Widloski, **U. Kamaci**, and F. Kamalabadi. *IEEE ICIP*. [link]
- **Efficient Sparsity-Based Inversion for Photon-Sieve Spectral Imagers with Transform Learning** 2017
  U. Kamaci, F. C. Akyon, T. Alkanat, and F. S. Oktem. *IEEE GlobalSIP*. [link]

Co-author of 5 further NASA ICON papers in *JGR Space Physics*, *Geophysical Research Letters*, and *Space Science Reviews*.

Poster Presentations

- **Toward Attaining the Diffraction Limit and Beyond in Astronomical Imaging** 2025
  F. Kamalabadi, **U. Kamaci**, and Z. Han. *BASP Frontiers Workshop*. [link]
- **Posterior Sampling for Solar Snapshot Spectroscopy with Score-Based Priors** 2025
  U. Kamaci and F. Kamalabadi. *BASP Frontiers Workshop*. [link]
- **Instantaneous Spectral Imaging with Learned Reconstruction** 2023
  U. Kamaci and F. Kamalabadi. *BASP Frontiers Workshop*. [link]

Honors

- **3rd Place**, METU EE STAR Undergraduate Research Competition (of 32 finalists) Mar 2017
- **Best Term Project**, EE230 Probability and Random Variables, METU Jun 2015
- **Dean's High Honor List** (8 semesters), METU 2013 – 2017